

model the hierarchical structure of datasets like the CPI, with each node in the network graph representing an RNN unit that captures the values of a specific (sub)-index. This architecture enables bidirectional information flow—allowing higher-level RNN nodes to influence lower-level ones and vice versa—through hierarchical informative priors applied to the RNNs’ parameters. This bidirectional exchange enhances predictions across all levels of the hierarchy.

The BiHRNN represents a substantial advancement over its predecessor, the Hierarchical Recurrent Neural Network (HRNN), which allowed information to flow only in one direction—from parent categories to child categories. By enabling bidirectional communication, the BiHRNN enhances predictive accuracy and reliability. Evaluation results reveal that the BiHRNN consistently outperforms both the HRNN and traditional methods, providing more precise and dependable inflation predictions. Across various metrics, such as prediction accuracy, overall accuracy, correlations, and goodness of fit, the BiHRNN demonstrates notable improvements over the HRNN, underscoring its superior ability to capture and model the hierarchical dependencies within the data.

The central aim of this thesis is to present the BiHRNN model, specifically designed to utilize hierarchical data structures for enhanced prediction accuracy. We demonstrate the model’s effectiveness using three distinct Consumer Price Index (CPI) datasets from Canada, Norway, and the US. In these cases, we forecast the CPI while accounting for multiple levels, ranging from broad economic indicators to specific categories, providing a comprehensive evaluation of the model’s performance across different hierarchical structures.

The thesis is structured as follows: Related work of existing methodologies in hierarchical data modeling and forecasting (pages 5-8), along with an overview of Recurrent Neural Networks (RNN) (page 9) and the previous Hierarchical Recurrent Neural Network (HRNN) model (pages 10-12). A detailed explanation of the BiHRNN model, including its formulation and inference mechanisms (pages 12-16). A description of the datasets used (pages 17-25), the baseline models (pages 26-27), evaluation metrics (page 28), and a thorough analysis of the BiHRNN model’s performance (pages 29-30). Interpretation of the results and the implications of the findings (pages 30-32), and potential directions for future work (pages 33).

2 Related Work

In this section, we examine existing research on inflation data modeling and forecasting, with an emphasis on methodologies that make use of the hierarchical structure of data to address complex challenges effectively.

2.1 *Time series*

Time series prediction plays a crucial role in various domains, such as energy management, supply chain optimization, sports analytics, and weather forecasting. It involves forecasting future values by analyzing previously observed data in a sequential order. Traditional approaches such as Autoregressive Integrated Moving Average (ARIMA) models have been commonly applied in inflation forecasting due to their straightforward nature and ease of interpretation [Box and Jenkins, 1970]. However, with the rise of deep learning techniques, more advanced models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have demonstrated substantial improvements in predictive accuracy [Hochreiter and Schmidhuber, 1997, Goodfellow et al., 2016]. Recent advancements in time series forecasting include the use of Transformers, as noted by [Zeng et al., 2022]. Transformers are highly effective at capturing complex patterns and long-range dependencies; however, their large number of parameters makes them susceptible to overfitting, particularly when applied to the smaller datasets often encountered in time series tasks. Additionally, their computational complexity, driven by the self-attention mechanism, can introduce unnecessary overhead, especially when simpler models often perform comparably for many forecasting tasks.

2.1.1 *Inflation Time-Series Prediction*

Forecasting inflation time series is critical for economic policy and decision-making. Traditionally, this has been achieved using econometric models such as Vector Autoregression (VAR) and structural models, which excel at capturing dynamic relationships between various economic indicators [Sims, 1980]. However, recent advancements in time series forecasting have spurred increasing interest in deep learning techniques for inflation prediction. Models like LSTM networks and hybrid approaches combining statistical and

machine learning methods have shown superior performance in capturing the complex and non-linear patterns inherent in inflation data [Cheng et al., 2019, Guo et al., 2021].

The Filtered Ensemble Wavelet Neural Network (FEWNet) stands out for its innovative approach to forecasting CPI inflation [Sengupta et al., 2024]. FEWNet utilizes wavelet transforms to decompose inflation data into high- and low-frequency components and integrates additional economic factors, such as economic policy uncertainty and geopolitical risk, to enhance forecasting accuracy. These transformed components, along with filtered exogenous variables, are processed through autoregressive neural networks to produce an ensemble forecast. By effectively capturing non-linearities and long-range dependencies through its adaptable architecture, FEWNet has demonstrated superior performance compared to traditional models, providing accurate forecasts and robust estimation of prediction uncertainty.

While some studies focus on machine learning techniques, other recent researches highlight the significance of trend and cross-sectional asymmetry measures in enhancing inflation forecasting methodologies. Trimmed-mean inflation estimators have demonstrated their effectiveness in predicting headline inflation for the Personal Consumption Expenditures (PCE) Price Index, substantially improving both point and density forecast accuracy over medium- and long-term horizons [Verbrugge and Zaman, 2024].

2.2 Hierarchical inflation Forecasting

Hierarchical inflation modeling structures inflation data into a hierarchy, where each disaggregated category, such as specific goods or services, contributes to the overall inflation index. This approach enhances prediction accuracy by leveraging the relationships and dependencies between different levels of the hierarchy, such as categories, subcategories, and aggregate indices. Forecasting within this framework often involves aggregating or disaggregating data to ensure consistency and coherence across all hierarchical levels, allowing for more precise and interpretable predictions.

An example of hierarchical inflation forecasting is the HRNN model [Barkan et al., 2023], the predecessor of BiHRNN. The HRNN model is specifically designed to address the unique challenges of inflation forecasting in datasets with hierarchical structures. Lower levels, such as specific categories of goods or regional inflation indices, often exhibit missing data and higher volatility in price changes compared to higher aggregate levels. By aligning with the principles of hierarchical Bayesian models, the HRNN assigns prior distributions to parameters at each level, capturing the relationships and dependencies within the hierarchy.

Hierarchical time series inflation models are particularly well-suited for inflation forecasting as they effectively integrate temporal dynamics and cross-level information. This capability is crucial for tasks like predicting the consumer price index, where capturing temporal trends, seasonality, and structural shifts in economic conditions is essential. For instance, fluctuations in specific product categories or regional price indices